



Introduction to Robotics

CSE 461

Class Topic: Introduction to Machine Learning and Neural Network

Tumor Size	Condition
10	malignant
2	benign
1	benign
5	malignant

Tumor Size	Condition
10	malignant
2	benign
1	benign
5	malignant



Tumor Size	Condition
10	malignant
2	benign
1	benign
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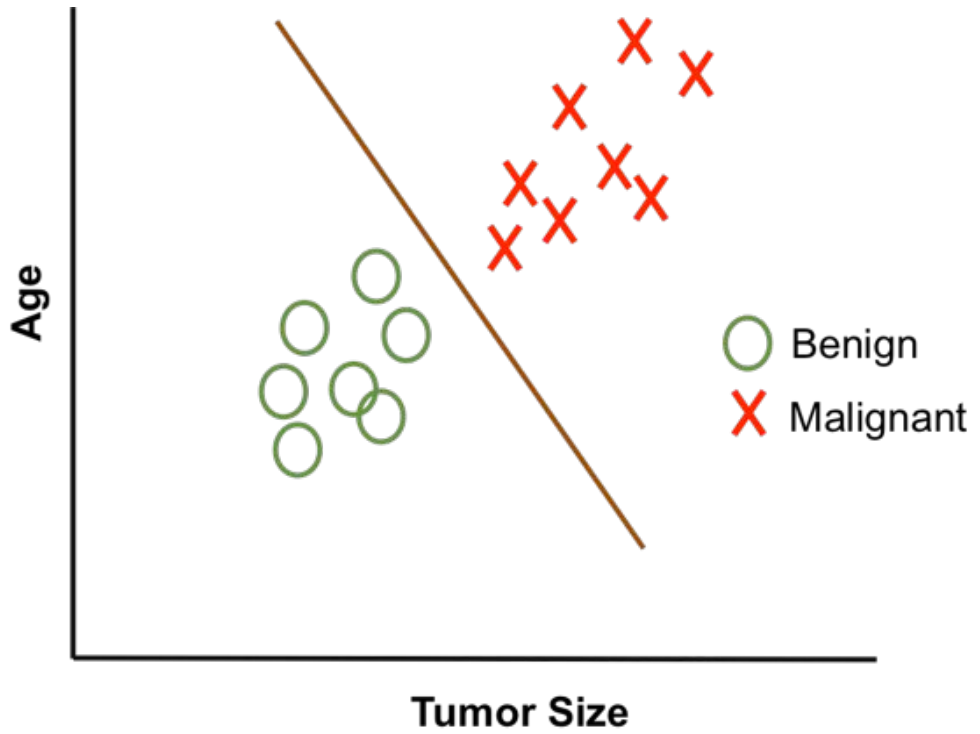


Tumor Size	Condition
10	malignant
2	benign
1	benign
5	malignant



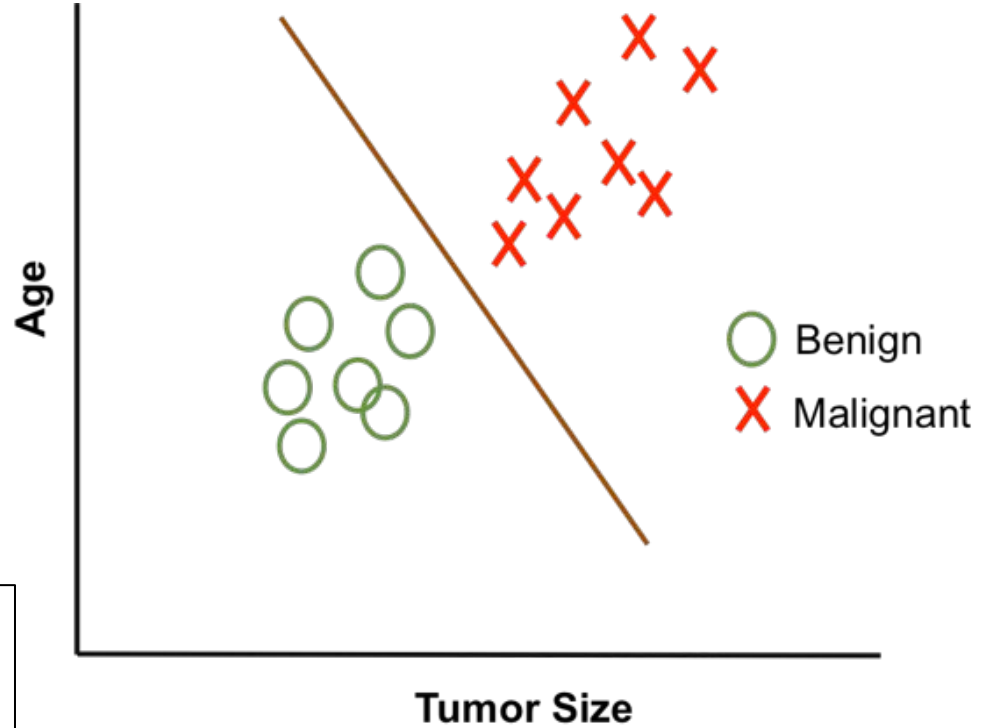
Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

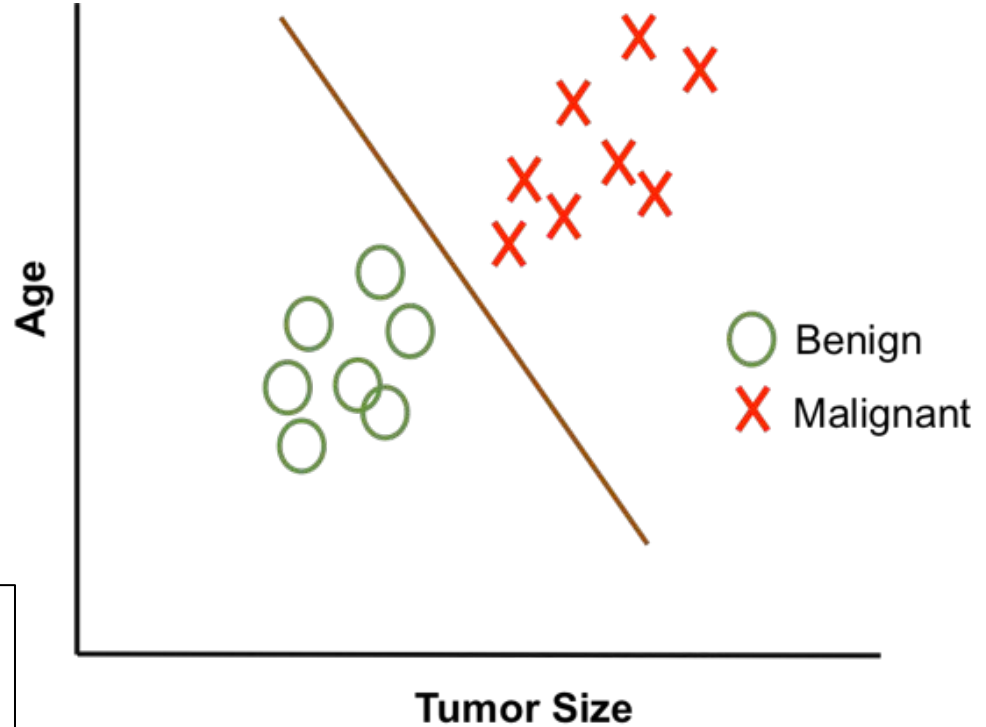


Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

$$0.9 * 10 + 0.1 * 50 = 9 + 5 = 14$$



Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant



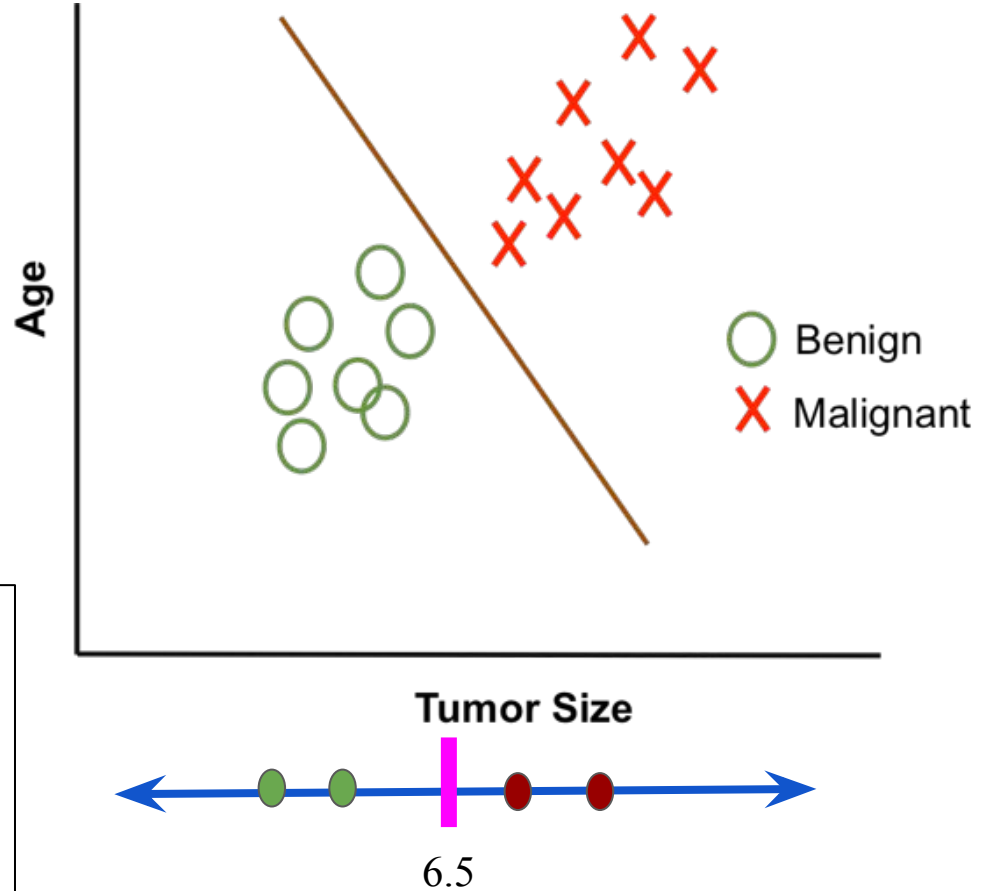
$$0.9*10 + 0.1*50 = 9 + 5 = 14$$

$$0.9*2 + 0.1*30 = 1.8 + 3 = 4.8$$

$$0.9*1 + 0.1*45 = 0.9 + 4.5 = 5.4$$

$$0.9*5 + 0.1*40 = 4.5 + 4 = 8.5$$

Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant



$$0.9 \cdot 10 + 0.1 \cdot 50 = 9 + 5 = 14$$

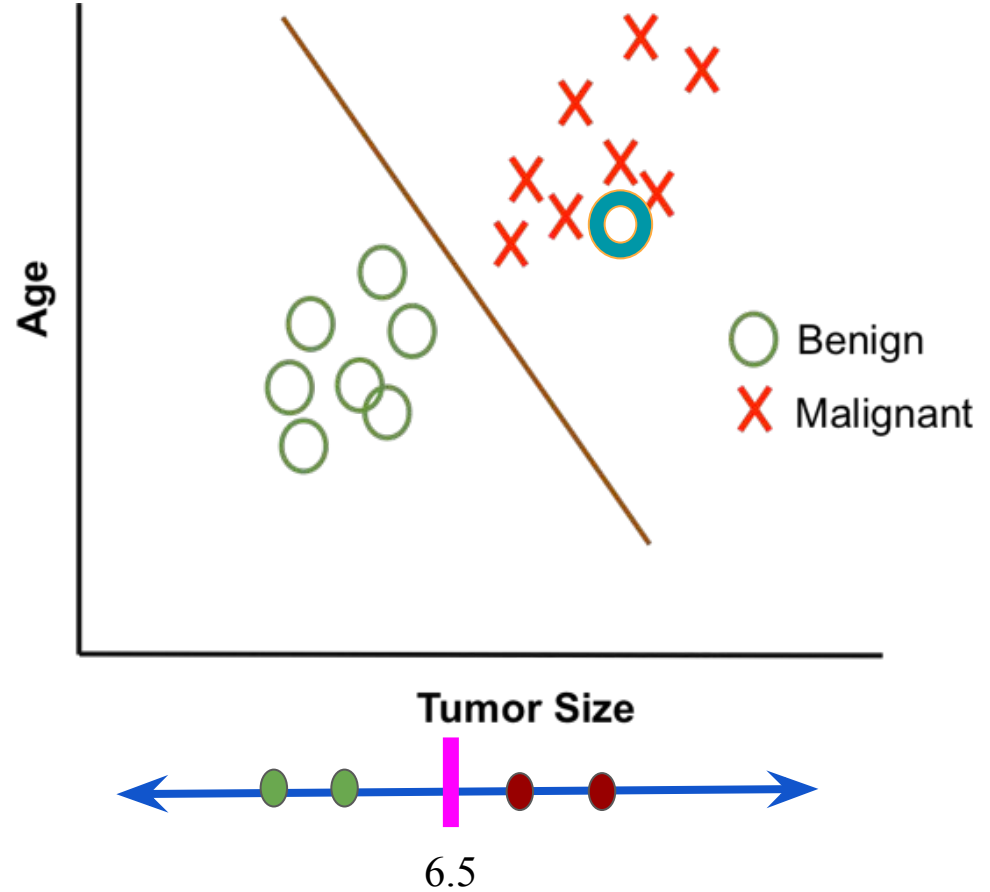
$$0.9 \cdot 2 + 0.1 \cdot 30 = 1.8 + 3 = 4.8$$

$$0.9 \cdot 1 + 0.1 \cdot 45 = 0.9 + 4.5 = 5.4$$

$$0.9 \cdot 5 + 0.1 \cdot 40 = 4.5 + 4 = 8.5$$

Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

Tumor Size = 9
Age = 42

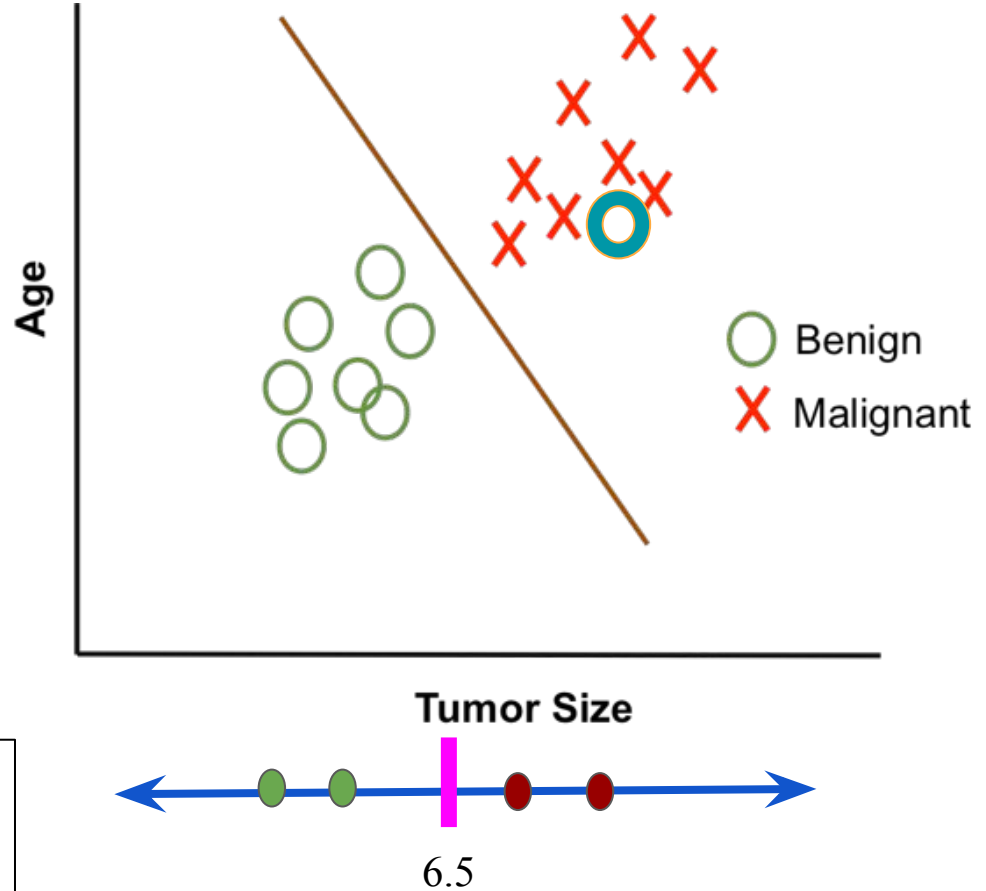


Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

Tumor Size = 9

Age = 42

$$0.9 \cdot 9 + 0.1 \cdot 42 = 8.1 + 4.2 = 12.3$$

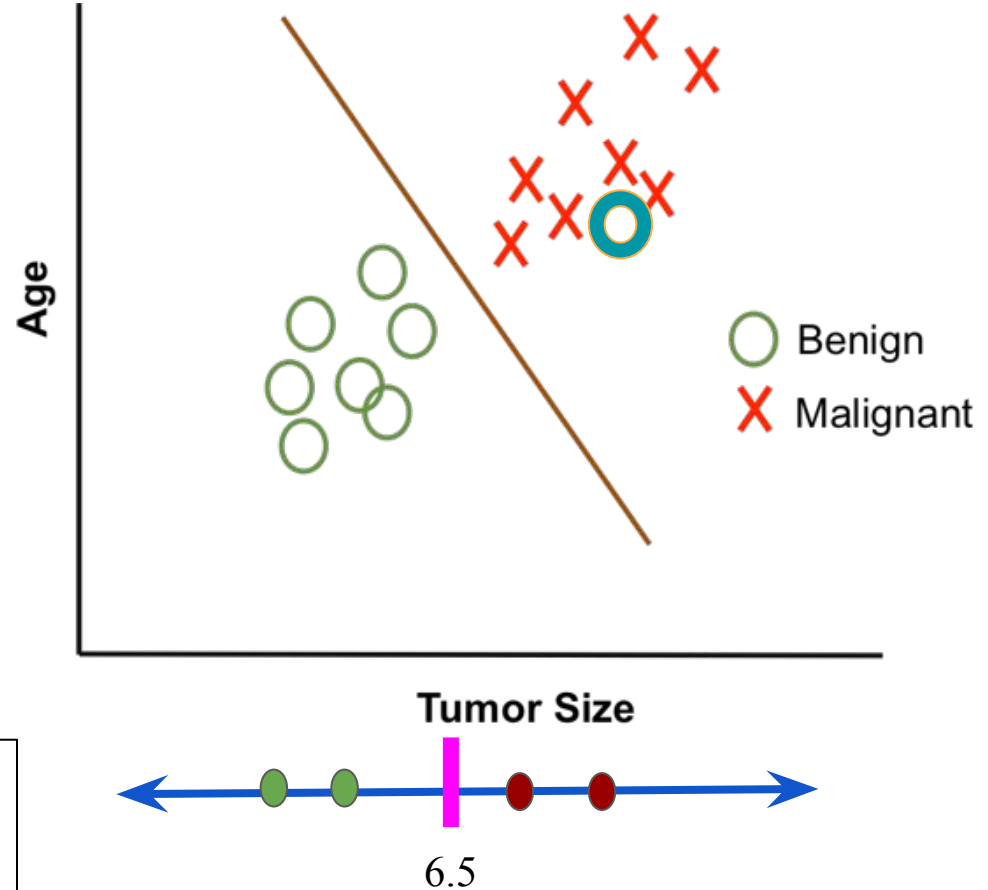


Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

Tumor Size = 9
Age = 42

$$0.9 \cdot 9 + 0.1 \cdot 42 = 8.1 + 4.2 = 12.3$$

malignant

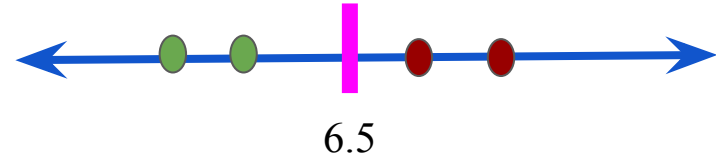


x1	x2	y
Tumor Size	Age	Condition
10	50	malignant
2	30	benign
1	45	benign
5	40	malignant

$$w1 * x1 + w2 * x2 = y$$

$$0.9 * 9 + 0.1 * 42 = 8.1 + 4.2 = 12.3$$

malignant



x1	x2	y
Tumor Size	Age	Condition
10	50	malignant
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1	45	benign
5	40	malignant

$$w1 * x1 + w2 * x2 = y$$

$x1, x2, \dots, xn$ = Features

$w1, w2, \dots, wn$ = Weights

y = output/ target

x1	x2	y
Tumor Size	Age	Condition
10	50	malignant
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1	45	benign
5	40	malignant

$$w1 * x1 + w2 * x2 = y$$

Let $w1 = 0.1$ and $w2 = 0.9$

$x1 = 2$ and $x2 = 5$

$$0.1 * 2 + 0.9 * 5 = 0.2 + 4.5 = 4.7$$

benign

x1	x2	y
Tumor Size	Age	Condition
10	50	malignant
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5	40	malignant

$$w1 * x1 + w2 * x2 = y$$

Let $w1 = 0.1$ and $w2 = 0.9$

$x1 = 2$ and $x2 = 30$

$$0.1 * 2 + 0.9 * 30 = 0.2 + 27 = 27.2$$

malignant

x1	x2	y
Tumor Size	Age	Condition
10	50	malignant
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$$w1 * x1 + w2 * x2 = y$$

Let $w1 = 0.1$ and $w2 = 0.9$
 $x1 = 2$ and $x2 = 30$
 $0.1 * 2 + 0.9 * 30 = 0.2 + 27 = 27.2$
malignant

Weights are very important.

Size(sq-feet)	Bedroom	Kitchen	Main road	Balcony	Price(dollar)
1600	4	1	0.2	2	20000
1400	3	1	0.5	1	12000
1600	2	1	0.2	2	10000
2000	4	2	0.1	3	43000

x1	x2	x3	x4	x4	y
Size(sq-feet)	Bedroom	Kitchen	Main road	Balcony	Price(dollar)
1600	4	1	0.2	2	20000
1400	3	1	0.5	1	12000
1600	2	1	0.2	2	10000
2000	4	2	0.1	3	43000

Example of Regression

x1	x2	x3	x4	x5	y
Size(sq-feet)	Bedroom	Kitchen	Main road	Balcony	Price(dollar)
1600	4	1	0.2	2	20000
1400	3	1	0.5	1	12000
1600	2	1	0.2	2	10000
2000	4	2	0.1	3	43000

$$w1 * x1 + w2 * x2 + + w4 * x4 = y$$

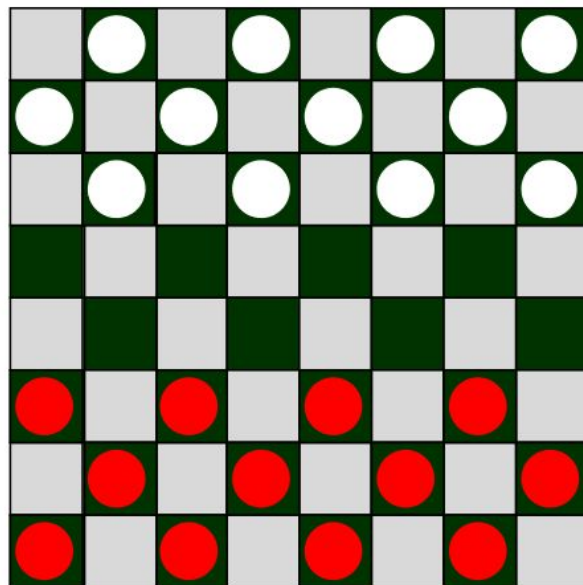
Two Types

1. Regression
2. Classification

Machine learning

"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



What if x_3 is not important?

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$

What if x_3 is not important?

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$

Simply set $w_3 = 0$

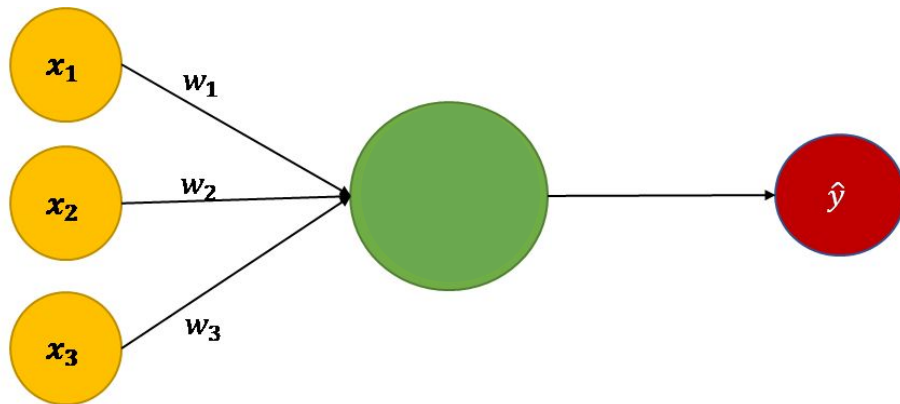
$$w_1 * x_1 + w_2 * x_2 = y$$

Mother Law of Machine Learning

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$

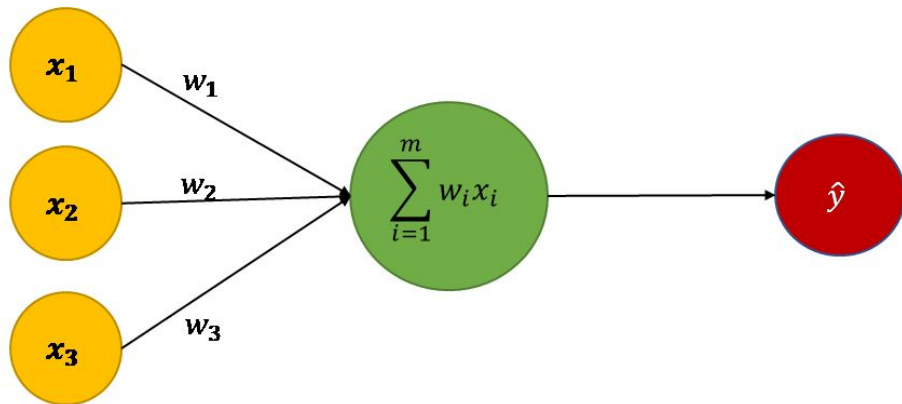
Mother Law of Machine Learning

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$



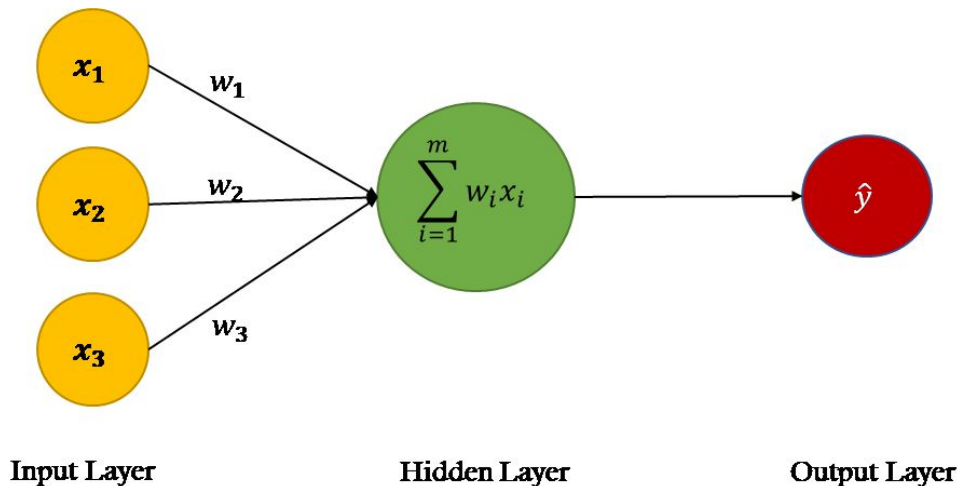
Mother Law of Machine Learning

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$



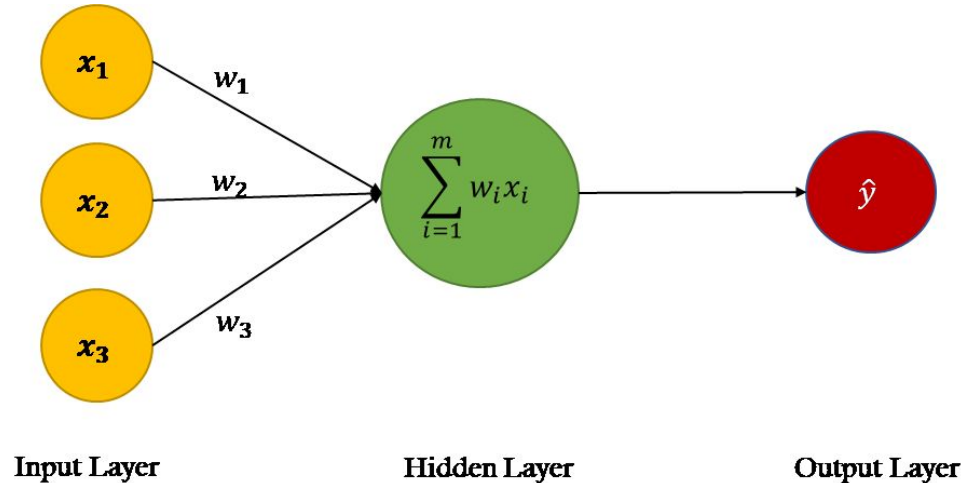
Mother Law of Machine Learning

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$

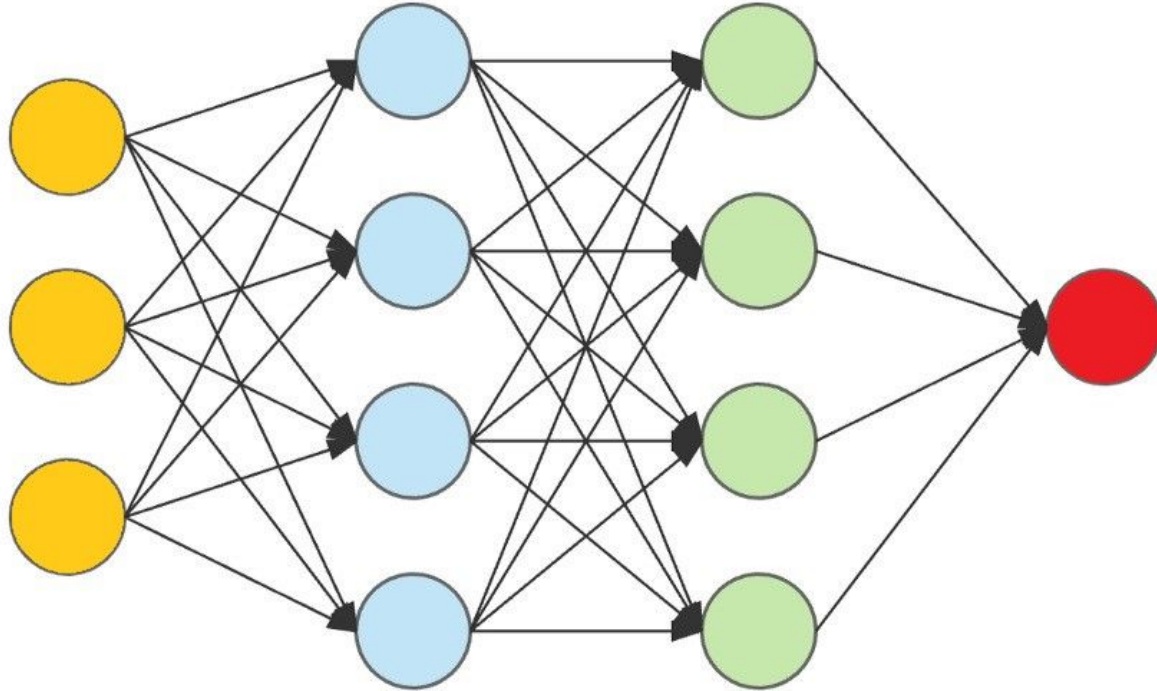


Neural Network

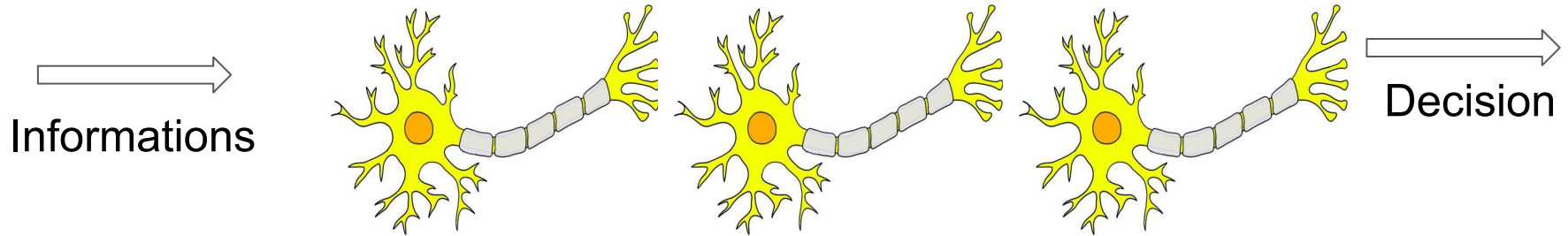
$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 = y$$



Neural Network

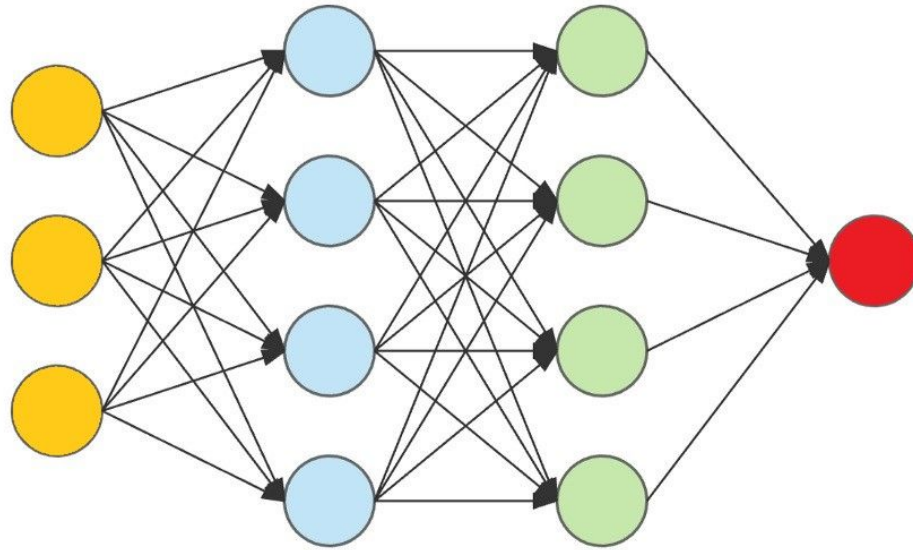


Human Brain



Machine Brain

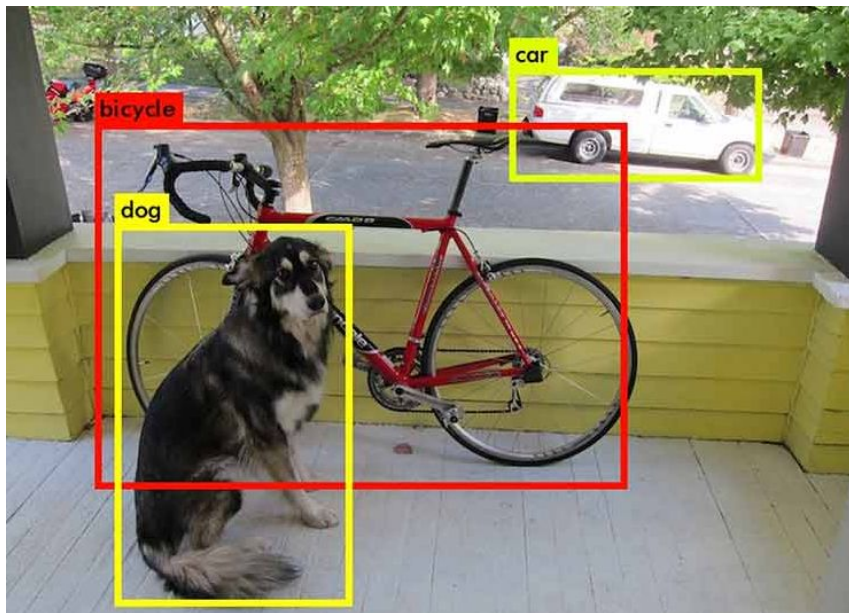
→
Informations



→
Decision

Next Class

CNN, Object Detection using Neural Network Learning



Thank You So Much